

BARNWELL, SC, USAWMO: **720602**Lat: **33.250N**Lon: **81.383W**Elev: **247**StdP: **14.57**Time Zone: **-5.00 (NAE)**Period: **09-19**WBAN: **00194****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	28.0	8.9	8.7	32.9	12.2	10.3	37.2	18.9	57.5	17.1	55.7	1.9	260	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	97.2	76.2	94.9	76.4	92.7	75.6	80.1	90.3	79.0	89.2	78.2	87.9	6.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	143.8	82.6	76.7	140.7	82.2	75.3	133.8	81.3	43.6	90.2	42.6	88.8	41.6	88.6	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	13.9	12.1	18.3	100.4	3.7	2.8	15.6	102.4	13.5	104.1	11.4	105.7	8.7	107.7		
(5)			16.8	81.3	3.7	1.0	14.1	82.0	11.9	82.6	9.8	83.1	7.1	83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.6	46.8	51.5	57.2	65.2	73.6	80.3	82.4	80.9	76.7	66.1	54.7	50.8
	DBStd	14.46	9.97	9.72	9.34	7.10	5.67	4.00	3.20	3.32	4.81	7.63	7.88	9.31
	HDD50	459	184	91	39	2	0	0	0	0	0	1	36	106
	HDD65	2182	569	389	275	85	10	0	0	0	1	79	323	451
	CDD50	6142	84	134	261	457	730	908	1004	957	800	501	176	130
	CDD65	2390	4	12	32	90	275	458	539	492	351	114	13	10
	CDH74	23638	20	85	330	1020	2664	4808	5736	4724	3115	1019	89	28
	CDH80	10576	1	6	64	300	1110	2394	2847	2201	1353	289	11	0
Wind	WSAvg	4.3	4.4	4.8	5.2	5.2	4.7	4.0	4.0	3.6	4.0	4.0	3.9	4.1
Precipitation	PrecAvg	47.0	4.1	3.7	4.2	2.8	3.0	5.5	5.2	5.5	3.8	3.5	2.6	3.3
	PrecMax	61.1	8.5	7.4	10.5	6.9	6.9	9.4	9.8	13.0	9.1	13.2	7.8	8.1
	PrecMin	37.6	1.7	1.3	0.7	0.9	0.9	1.8	2.8	2.9	0.7	0.0	0.4	0.5
	PrecStd	7.4	1.9	1.6	2.2	1.7	1.5	2.1	1.7	2.5	1.9	3.1	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.0	81.2	84.3	90.1	97.1	99.4	99.8	99.0	95.3	90.6	81.4	77.4
		MCWB	65.4	66.8	64.6	69.5	71.3	77.5	77.3	77.1	74.2	75.6	69.0	70.4
	2%	DB	72.1	76.7	81.6	86.4	91.4	96.9	97.2	95.1	91.5	86.2	75.5	73.1
		MCWB	62.8	64.3	65.0	67.5	71.0	76.4	77.0	77.2	74.1	72.3	64.0	66.8
	5%	DB	67.7	72.7	78.6	83.6	90.0	93.2	94.9	92.6	90.4	82.4	72.7	70.2
		MCWB	61.1	61.7	63.1	65.7	70.7	75.3	77.4	76.2	73.7	70.0	63.0	65.3
	10%	DB	63.8	68.3	73.1	81.0	86.3	91.0	92.5	90.5	88.3	80.6	70.1	66.1
		MCWB	58.0	60.5	60.3	65.3	70.7	75.1	77.1	76.0	73.5	68.6	63.0	62.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.2	69.9	69.8	74.6	76.9	81.7	81.2	81.0	78.9	78.7	72.4	73.6
		MCDB	71.6	75.3	78.3	83.3	85.4	90.3	92.2	90.8	88.7	84.3	75.9	74.8
	2%	WB	65.5	67.5	68.0	71.8	74.9	79.3	80.1	79.5	77.5	76.5	69.7	69.9
		MCDB	71.0	74.1	76.0	79.7	83.3	89.4	91.7	88.8	86.6	81.7	72.9	71.4
	5%	WB	63.1	64.8	66.1	69.9	73.6	78.1	79.0	78.6	76.5	73.9	67.0	66.2
		MCDB	65.6	70.2	73.8	77.2	82.3	88.9	90.5	87.5	84.9	79.7	70.5	68.8
	10%	WB	59.8	62.2	63.7	68.0	72.6	77.0	78.1	77.8	75.4	72.1	63.8	63.2
		MCDB	63.2	66.9	69.7	75.6	81.3	87.7	88.6	86.4	83.2	77.3	67.9	65.0
Mean Daily Temperature Range	5% DB	MDBR	21.6	22.7	24.1	24.4	21.9	20.9	19.6	18.8	20.4	23.3	23.3	20.2
		MCDBR	25.0	25.9	27.8	29.5	26.9	23.9	23.1	21.8	23.8	24.4	26.6	22.2
		MCWBR	16.3	14.9	13.1	11.7	8.9	7.6	6.9	6.7	7.9	11.3	15.9	15.0
	5% WB	MCDBR	19.9	21.2	22.3	20.4	19.7	21.2	20.1	18.5	19.2	18.3	19.7	17.5
		MCWBR	16.1	14.5	11.4	10.4	7.9	7.6	6.7	6.5	7.2	9.6	14.7	13.9
Clear-Sky Solar Irradiance	taub		0.325	0.333	0.368	0.404	0.470	0.509	0.551	0.529	0.437	0.401	0.348	0.322
	taud		2.488	2.463	2.395	2.287	2.143	2.082	1.986	2.039	2.283	2.345	2.462	2.536
	Ebn at Noon		283	292	288	281	262	251	240	243	263	264	271	278
	Edh at Noon		28	31	36	42	49	52	57	53	40	35	28	25
All-Sky Solar Radiation	RadAvg		905	1104	1445	1822	1964	2007	1932	1760	1519	1286	991	789
	RadStd		46	110	109	126	140	141	135	93	128	149	74	72